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DETAIL SPECIFICATION

POTASSIUM SULFATE (FOR ORDNANCE USE)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements, examinations, and tests for potassium sulfate intended for ordnance applications.

1.2 Classification. Potassium sulfate furnished under this specification are of the following types:

Type I	for use in the manufacture of propellants
Type II	for use in flash-reducing mixtures
Type III	as a filter in thermosetting resin potting applications
Type IV	for use in ballistic salt rods

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 or 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in section 3 or 4 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to: U.S. Army DEVCOM Armaments Center, ATTN: FCDD-ACE-QSA, Picatinny Arsenal, New Jersey 07806-5000 or emailed to usarmy.pica.devcom-ac.list.ardec-stdzn-branch@army.mil. Since contact information can change, you may want to verify the currency of this information using ASSIST Online database at https://assist.dla.mil.

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-650	-	Explosive; Sampling, Inspection and Testing
MIL-STD-1916	-	DOD Preferred Methods for Acceptance of Product

(Copies of these documents are available online at https://quicksearch.dla.mil/).

2.3 Non-Government standards and publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM B330	-	Standard Test Methods for Estimating Average Particle Size of Metal Powders and Related Compounds Using Air Permeability
ASTM D1193	-	Standard Specification for Reagent Water
ASTM E11	-	Standard Specification for Woven Wire Test Sieve Cloth and Test Sieves
ASTM E70	-	Standard Test Method for pH of Aqueous Solutions with the Glass Electrode
ASTM E288	-	Standard Specification for Laboratory Glass Volumetric Flasks
ASTM E300	-	Standard Practice for Sampling Industrial Chemicals
ASTM E969	-	Standard Specification for Glass Volumetric (Transfer) Pipets

(Copies of these documents are available from https://www.astm.org/.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Required inspections.

3.1.1 First article inspection. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

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3.1.2 Conformance. A sample shall be subject to conformance inspection in accordance with 4.3.

3.2 Chemical properties. Potassium sulfate Types I and II shall conform to the chemical requirements specified in Table I.

TABLE I. Chemical requirements for potassium sulfate.

Requirement	Types I & II
Purity (moisture-free basis) Based on sulfate content, percent minimum Based on potassium content, percent minimum	 99.0 99.0
Moisture, weight percent maximum	1.0
Insoluble matter, weight percent maximum	0.1
Grit	None
pH	7 ± 1
Chlorides (as potassium chloride), weight percent maximum	0.02

3.3 Granulation requirements. Potassium sulfate shall comply with the granulation requirements specified in Table II.

TABLE II. Granulation requirements.

U.S Alternate Sieve Number	Type I		Type II		Type III		Type IV	
	Min	Max	Min	Max	Min	Max	Min	Max
% Through No. 4	-	-	100.0	-	-	-	-	-
% Retained on No. 30	-	-	30	-	-	-	-	-
% Through No. 50	100.0	-	-	5	-	-	-	-
% Retained on No. 70	-	5	-	-	-	-	-	5
% Through No. 100	-	-	-	-	-	-	90	-
% Through No. 200	-	-	-	-	100.0	-	-	-
% Retained on No. 325	-	-	-	-	-	20	-	-

3.4 Average particle size. The average particle size for Type III shall be not less than 0.5 microns.

3.5 Workmanship. Potassium sulfate shall be uniform and free from visible impurities and foreign material such as dirt, rust, paint, wooden particles, or any other foreign objects.

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4. VERIFICATION

TABLE III. Requirements/verification cross reference matrix.

Method of verification 1 – Analysis 2 – Demonstration 3 – Examination 4 – Test						Classes of verification A – First article B – Conformance		
Section 3 Requirement	Description	Verification Methods				Verification Class		Section 4 Verification
		1	2	3	4	A	B	
3.1.1	First article	X		X	X	X		4.2
3.1.2	Conformance	X		X	X		X	4.3
3.2	Moisture	X			X	X	X	4.4.2
3.2	Purity	X			X	X	X	4.4.3
3.2	Insoluble matter	X			X	X	X	4.4.4
3.2	Grit	X			X	X	X	4.4.4
3.2	pH	X			X	X	X	4.4.5
3.2	Chlorides	X			X	X	X	4.4.6
3.3	Granulation	X			X	X	X	4.4.7
3.4	Average particle size	X			X	X	X	4.4.8
3.5	Workmanship			X	X	X	X	4.4.9

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see paragraph 4.2).

b. Conformance inspection (see paragraph 4.3).

4.2 First article inspection. When specified (see 6.2), a sample shall be subjected to first article verification in accordance with Table IV. Samples of not less than 450 grams (g) of potassium sulfate collected in accordance with 4.3.2 shall be submitted for examination. Samples shall be produced by the contractor using the same process procedures, material supply sources, production equipment, and production line as will be used in regular production (see 4.3.2).

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TABLE IV. First article acceptance tests.

Category	Examination or Test	Requirement Paragraph	Inspection Method
Critical	None defined		
Major			
101	Moisture	3.2	4.4.2
102	Purity	3.2	4.4.3
103	Insoluble matter	3.2	4.4.4
104	Grit	3.2	4.4.4
105	pH	3.2	4.4.5
106	Chlorides	3.2	4.4.6
107	Granulation	3.3	4.4.7
108	Average particle size (Type III only)	3.4	4.4.8
Minor			
201	Workmanship	3.5	4.4.9

4.2.1 Rejection. If any sample fails to comply with any of the applicable requirements the first article quantity shall be rejected.

4.3 Conformance inspection.

4.3.1 Inspection lot formation. Inspection lots and batches shall comply with the MIL-STD-1916, paragraph "Formation and identification of lots or batches." A lot shall consist of a single batch or multiple batches that have been blended to a uniform consistency of potassium sulfate produced by one manufacturer in accordance meeting all requirements of this specification and revision under one continuous set of operating conditions. Each batch shall consist of a quantity of potassium sulfate that has been subjected to the same chemical and physical process intended to make the final product homogeneous.

4.3.2 Sampling. Sampling for inspection shall be performed at either the supplier/distributor location where the potassium sulfate is packed, or at the manufacturer's receiving facility where the material is unpacked for use in the next higher sub-assembly or formulation.

4.3.2.1 Sampling at the manufacturer. Samples shall be collected from the beginning, middle, and end of every batch in every lot in accordance with ASTM E300. A composite sample of not less than 450 g shall be formed which is assumed to be representative from a batch. All composite samples shall be verified in accordance with 4.3.3. If any sample fails to comply with any applicable requirements the batch shall be rejected.

4.3.2.2 Sampling at the receiving facility. Potassium sulfate batches shipped in containers (e.g., drums, totes, boxes, etc.) shall be sampled using Verification Level I of MIL-STD-1916, where

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one unit is defined as one container. Each container shall be sampled in accordance with ASTM E300 using a sample thief. Approximately equal portions shall be taken from each selected container. This material shall be placed in a secondary container and mixed to form a composite sample consisting of a minimum of 450 g. The composite sample shall be tested and verified in accordance with 4.3.3. If any sample fails to comply with any applicable test requirements the lot represented by the sample shall be rejected.

4.3.3 Potassium sulfate.			
Category	Examination or Test	Requirement Paragraph	Inspection Method
Critical	None defined		
Major			
101	Moisture	3.2	4.4.2
102	Purity	3.2	4.4.3
103	Insoluble matter	3.2	4.4.4
104	Grit	3.2	4.4.4
105	pH	3.2	4.4.5
106	Chlorides	3.2	4.4.6
107	Granulation	3.3	4.4.7
108	Average particle size (Type III only)	3.4	4.4.8
Minor			
201	Workmanship	3.5	4.4.9

4.4 Test procedures.

4.4.1 Materials and equipment.

a. Water, deionized (DIW) or distilled, conforming to ASTM D1193 Type I or II; all subsequent mentions of water refer hereto.

b. Volumetric flasks, Class A per ASTM E288; all subsequent mentions of volumetric flasks refer hereto

4.4.2 Determination of moisture. An accurately weighed portion of 5.0 ± 0.1 g of the specimen shall be transferred to a tared moisture dish and the gross weight recorded to the nearest 0.0001g. The dish and contents shall be heated at 100° to 105°C for 2 hours, cooled in a desiccator to room temperature and weighed. The loss in weight shall be calculated as a percentage of moisture in the sample.

$$Percent\ moisture = \frac{W_w - W_d}{W_w} * 100$$

Where

W_w = Weight of wet sample

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W_d = Weight of dry sample

4.4.3 Determination of purity. Samples shall be tested for purity based on both sulfate content (via either 4.4.3.1 or 4.4.3.2) and potassium content (via either 4.4.3.3 or 4.4.3.4). The sample shall comply with both sulfate content and potassium content requirements (see 3.2) to be compliant with purity requirements.

4.4.3.1 Based on sulfate content.

4.4.3.1.1 Preparation of the ion-exchange column. Into a 100 milliliters (mL) beaker containing 50 mL of water add 50 g of Dowex 50W X8, H + (20-50 mesh) cation exchange resin or equivalent and allow to stand for 1 hour until the swelling of resin particles ceases. Transfer the resin to a chromatographic tube 3 centimeter (cm) inner diameter x 38 cm length provided with a perforated plate and Teflon stopcock. The resin bed should rest on a pad of glass wool supported by the perforated plate and be 8 cm high. If the resin is in the sodium form, it shall be converted to the acid form by passing 500 mL of 6 molar (M) hydrochloric acid through the column and then washing with water until the effluent shows a pH of 5 or higher (approximately 200 mL). A 0.5 g specimen of potassium sulfate shall be dissolved in 50 mL of water and transferred quantitatively to the ion-exchange. The solution shall be drained through the column at a rate not exceeding 35 mL per minute. While draining the solution from the column, the liquid level should never be allowed to fall below the top of the resin bed. The column shall then be washed with four 50 mL portions of water. The column is then ready for use. The resin shall be kept covered with water when the column is not is use. Thus prepared, the column can be used in the analysis of seven 0.5 g samples of potassium sulfate before regeneration is necessary (see 4.4.3.1.2).

4.4.3.1.2 Regeneration of the ion-exchange columns. To regenerate the ion-exchange resin pass 50 mL of 6 M hydrochloric acid through the column and follow with water washings until the pH of the effluent water is 5 or higher. Remove the ion-exchange resin from the column and slurry with water. Replace the resin in the column. Dissolve 0.5 g of potassium sulfate in 50 mL of water and pass the solution through the ion-exchange column at a rate not exceeding 35 mL per minute. Wash the column with four 50 mL portions of water taking care to stop the flow of the final wash while the resin is still covered with water. The regenerated column can be used in the analysis of 7 additional 0.5 g samples of potassium sulfate before regeneration is again necessary.

4.4.3.1.3 Sulfate content procedure. Accurately weigh 0.5000 ± 0.0100 g of potassium sulfate to the nearest 0.0001 g and transfer to a clean 250 mL beaker. Add 50 mL of water and swirl gently until all of the sample has dissolved. Transfer the potassium sulfate solution to the resin column and allow it to pass through the column, at a rate not exceeding 35 mL per minute, and into a clean 600 mL beaker. Rinse down the sides of the 250 mL beaker with 50 mL of water. Transfer the rinse water to the resin column when the last portion of the potassium sulfate solution has almost reached the upper level of the resin column. Rinse the column with 3 additional 50 mL water rinses stopping the flow through the column before the last portion of the final rinse reaches the level of the resin. The water rinses shall also be caught in the 600 mL beaker. Heat to boiling on a hot plate to expel any carbon dioxide. Add 50 mL of 2 percent barium chloride solution (10 g of barium chloride dihydrate in 500 mL of water) dropwise to the heated solution in the 600 mL beaker, over a period of 5 minutes, with moderate stirring. Allow the precipitate to settle out and add 5 more drops of barium

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chloride solution to the clear solution over the precipitate to see if precipitation has been completed. If precipitation has not been completed, add additional barium chloride solution until no further precipitation occurs. Cover the beaker and contents with a watch glass, place it on a steam bath and allow it to digest for not less than 3 hours. Let the beaker and contents stand overnight. Ignite a fine porosity porcelain crucible to constant weight (± 0.2 mg) at 600 ± 50 °C. Decant most of the supernatant solution through using vacuum filtration. If any precipitate passes through the filtering crucible it shall be poured back into the crucible repeatedly until the filtrate is clear. After most of the supernatant liquid has passed through the filter, quantitatively transfer the remaining precipitate to the filtering crucible with the aid of a rubber policeman. Dislodge and transfer any precipitate adhering to the beaker, stirring rod and rubber policeman with the aid of a small piece of ashless filter paper and a stream of water from a squirt bottle. Place the filter paper in the crucible containing the precipitate. Wash the precipitate with water until the water washing gives a negative chloride test with silver nitrate solution. Dry the crucible and contents in an oven set at 100 ± 5°C for 30 minutes and then transfer to a muffle furnace set at 600 ± 50°C and ignite for 1 hour. Cool the crucible and contents in a desiccator and weigh to the nearest 0.0001 g. Return the crucible to the furnace for 10 minutes, cool in the desiccator and weigh. Repeat this procedure until 2 consecutive weighings agree within 0.0002g. Calculate the percent potassium sulfate as follows:

$$\text{Potassium sulfate, percent} = \frac{74.65*A}{W}$$

where:

A = weight of barium sulfate precipitate in g.
W = sample weight in g corrected for moisture.

4.4.3.2 Based on sulfate content, alternate method by ion chromatography (IC).

4.4.3.2.1 Reagents. All chemicals used shall be reagent grade, or as specified below.

a. Sulfate reference: NIST-traceable sulfate standard, 1000 ± 5 mg/L at 95% confidence.
b. Eluent, per IC manufacturer recommendations, capable of eluting sulfate within a reasonable time while resolving its peak from possible contaminants, e.g. an aqueous solution containing 2 to 10 millimolar sodium carbonate and 1 to 3 millimolar sodium bicarbonate.

4.4.3.2.2 Equipment.

a. Analytical balance, 200 gram capacity, 0.0001 gram resolution or better.
b. Oven, set to 100 - 105 °C
c. Ion chromatograph with a fixed-volume loop injector, anion suppression, and conductivity detection.
d. Ion chromatography column, typically containing a quaternary-amine based stationary phase, which together with the proper eluent can baseline-resolve sulfate and completely elute it in a reasonable time (e.g. within twenty minutes or less).
e. Volumetric flasks, Class A per ASTM E288

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f. Volumetric pipettes, “To deliver”: Class A per ASTM E969, calibrated within three months of use, and verified to have ± 0.1% relative precision (or better) at 95% confidence, i.e. typically pipettes larger than or equal to 3 mL.

4.4.3.2.3 Calibration standards. Standards shall be prepared volumetrically. The following procedure is recommended but may be modified to allow for variation in instrumental set-up. At least four working standards are recommended to check response linearity or to warrant the use of a non-linear calibration.

a. Prepare an intermediate 100 mg/L sulfate standard.
b. Accurately dilute a 50 mL aliquot of the 1000mg/L NIST-traceable stock solution to 500 mL with water. Mix well.
c. Calculate the exact concentration based on the 50 mL pipette calibration and on the certified stock concentration.
d. Prepare working standards at concentrations 8-12 mg/L.
e. Accurately dilute each of 8 mL, 9 mL, 10 mL, and 12 mL (e.g. 2 x 6 mL) aliquots of the intermediate 100 mg/L sulfate standard to 100 mL with water. Mix well.
f. Calculate the exact concentrations in mg/L based on the calibrations of the respective pipettes and previously calculated concentration of the 100 mg/L sulfate standard.
g. Transfer the working standards to instrument sampling vials.

4.4.3.2.4 Sample preparation. The analysis shall be performed on two replicate portions, 0.36 gram each, of potassium sulfate, which had been previously dried for two hours at 100-105°C. Alternately, the raw sample weight shall be corrected for moisture based on the results from moisture testing.

a. Weigh 0.3600 ± 0.0050 g potassium sulfate sample to the nearest 0.1 mg and transfer quantitatively to a 1 L volumetric flask.
b. Add enough water to dissolve, then make up to volume with water and mix well. Prepare a duplicate in the same manner.
c. Accurately dilute a 10 mL aliquot of each replicate to 200 mL with water. Mix well.
d. Calculate the exact concentration “S” in mg/L of sample in the 200 mL flasks based on the calibration of the 10 mL pipette(s) and sample weights used.
e. Transfer the solutions to instrument sampling vials.

4.4.3.2.5 Procedure. Start up the ion chromatograph in accordance with manufacturer instructions. Set up a method appropriate for separation of common anions such as chloride, nitrate, phosphate, and sulfate; refer to column manufacturer’s recommendations. The instrument shall be equilibrated until a steady, linear baseline is obtained. It is recommended to inject 1-3 samples of water while the instrument is equilibrating to make sure the chromatograms do not contain any analytes other than water and/or system peaks. Inject each of the standards in order of increasing concentration. Ensure that the sulfate peak completely elutes during each run. Inject each of the two dilute samples. It is recommended to make replicate injections to verify repeatability. Identify the sulfate peak, which elutes at the end of each run. For both standards and samples, it shall be the largest peak in the chromatogram, and its retention time shall not vary by more than 5%. Integrate each chromatogram and obtain the sulfate peak areas. From the working standards, construct a

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calibration curve of sulfate concentration versus sulfate peak area. The curve can be linear or quadratic and shall have $R^2 \geq 0.9999$. From the calibration curve, calculate sulfate concentrations "C" for each of the two samples. Calculate purity based on sulfate as follows:

$$\text{Purity based on sulfate} = \frac{181.4 * C}{S}$$

where:

C = calculated sulfate concentration of dilute sample, mg/L
S = calculated sample concentration of dilute sample, mg/L

Calculate and report the average of the two purity values.

4.4.3.3 Based on potassium content.

4.4.3.3.1 Reagents. All chemicals used shall be reagent grade, or as specified below.

a. Acetic acid, 10% v/v. Ten (10) mL of glacial acetic acid shall be diluted to 100 mL with water.

b. Acetic acid, 0.5% v/v. Half (0.5) of a mL of glacial acetic acid shall be diluted to 100 mL with water.

c. Sodium acetate, 50% w/v. Fifty (50) g of sodium acetate trihydrate shall be dissolved in water and diluted to 100 mL.

d. Aluminum chloride, 0.2M. Five (5) g of aluminum chloride hexahydrate, $AlCl_3 \cdot 6H_2O$, shall be dissolved in 50 mL of water, and diluted to 100 mL.

e. Sodium tetraphenylboron, approximately 0.1M. Thirty-four (34) g of sodium tetraphenylboron shall be dissolved in 900 mL of water and diluted to 1.00 ± 0.01 L. Four (4) mL of 0.2M aluminum chloride solution shall be added. The solution shall be mixed and allowed to stand for at least 30 min, swirling occasionally until most of the solid dissolves. The solution shall be filtered through Whatman No. 40 filter paper and stored in a polyethylene bottle. It shall be re-filtered before use if any turbidity or suspended matter is present but shall not be used if older than 48 hours.

f. Potassium tetraphenylboron, saturated wash solution. A quantity of potassium tetraphenylboron shall be prepared from reagent grade potassium sulfate using the procedure prescribed for sample analysis (4.4.3.3.2) except that the precipitate can be collected on a 5-cm fritted Buchner funnel and shall be washed with 50 mL of 0.5% acetic acid. A 0.35g portion of this potassium tetraphenylboron shall be shaken for two (2) hours with 300 mL of water in a glass stoppered flask, and the resulting solution filtered through a No. 40 Whatman filter paper.

4.4.3.3.2 Potassium content procedure. An accurately weighed portion of 0.2200 ± 0.0050 g of potassium sulfate shall be transferred to a 250 mL beaker and dissolved in 100 mL of water. The solution shall be adjusted, if necessary, to a pH of 5 to 6 with 10% acetic acid or 50% sodium acetate. A 0.2 mL aliquot of 0.2M aluminum chloride shall be added. The solution shall be mixed well and heated to 70°C. Fifty (50) mL of 0.1M sodium tetraphenylboron shall be added dropwise with constant stirring. The suspension shall be cooled to room temperature and allowed to stand for 5 minutes. The suspension shall be filtered, and the precipitate collected on a tared, medium porosity glass filtering crucible. The beaker, crucible and precipitate shall be washed with five (5) x 10 mL

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portions of the saturated potassium tetraphenylboron wash solution. The crucible and contents shall be dried at 120 ± 5°C to constant weight, cooled in a desiccator, and weighed. The purity is calculated as follows:

$$\text{Purity, percent} = 24.316 * \frac{A}{W}$$

where:

A = weight of dried precipitate corrected for blank, g
W = weight of sample corrected for moisture, g

At least two replicate samples and a blank shall be analyzed to ensure accuracy and reproducibility. The blank weight shall be subtracted from each precipitate weight before calculating purity. Additionally, the standard deviation of replicates shall be no more than 0.5% wt, otherwise a total of five (5) replicates shall be run and the average reported.

4.4.3.4 Based on potassium content (alternate method). Use a 0.40-0.42 g sample weighed to the nearest 0.0001g and proceed in the manner used for sulfate-based purity analysis in 4.4.3.1 but stop after the final column rinse has been collected in 4.4.3.1.3 (before boiling). Add five (5) drops of 1% solution of phenolphthalein in alcohol and titrate with a standard 0.1 normal (N) solution of sodium hydroxide to a very light pink that lasts at least thirty seconds. Run a blank determination on an equal amount of water. Calculate potassium-based purity as follows:

$$\text{Potassium sulfate percent by weight} = \frac{8.713 * V * N}{W}$$

where:

V = blank-corrected volume of 0.1N sodium hydroxide, mL
N = effective concentration of 0.1N sodium hydroxide, mol/L
W = sample weight corrected for moisture, grams

Note: The titrated solution may be further used for sulfate-based purity in 4.4.3.1.3, however, add one and one half (1.5) mL of concentrated hydrochloric acid prior to boiling.

4.4.4 Determination of insoluble matter and grit. A weighed portion of 20 ± 0.1 g of potassium sulfate shall be dissolved in 200 ± 10 mL of hot (80 - 100°C) water in a beaker. The solution shall be filtered through a tared, fine porosity filtering crucible. The crucible and residue shall be washed with hot water until the washing no longer gives a positive test for sulfate when tested with 1mL of approximately 10% barium chloride solution. The crucible and contents shall be dried at 100° to 105°C for 1 hour, cooled in a desiccator and weighed. The increase in weight shall be calculated as percentage of insoluble matter in the sample. The material retained in the crucible shall be transferred to a smooth glass slide and the material shall be pressed and rubbed between glass slides determining the presence of grit by scratching noise and scratches on the glass slides.

4.4.5 pH. Weigh 5 ± 0.05 g of sample and place in a 250 mL beaker. Add 100 mL of freshly boiled and cooled water to the beaker and dissolve the sample. Determine the pH of this solution in accordance with ASTM E70.

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4.4.6 Determination of chlorides (as potassium chloride).

4.4.6.1 Determination of chlorides (as potassium chloride). A weighed portion of 50.0 ± 0.5 g of potassium sulfate shall be transferred to a beaker, dissolved in 400 mL of water, and the solution filtered. Twenty (20) mL of 5 percent nitric acid solution and 5 mL of 5 percent silver nitrate solution shall be added to the filtrate. The solution shall be heated to boiling and agitated vigorously to coagulate the precipitate of silver chloride. Exposure of the precipitate to strong light shall be avoided, as strong light causes the exposed surface to decompose and liberate chlorine. The crucible shall be washed with 1 percent nitric acid and heated at 140° to 150°C for 1 hour and cooled in a desiccator before taring. The solution shall be filtered through the tared filtering crucible and the residue washed with 1 percent nitric acid. The crucible and silver chloride residue shall be dried at 140° to 150°C for 2 hours, then cooled in the desiccator for 1 hour.

$$\text{Percentage of potassium chloride} = \frac{52.02 * A}{W}$$

where:

A = weight of silver chloride residue, g
W = weight of specimen, g

4.4.6.2 Determination of chlorides (as potassium chloride), alternate method.

4.4.6.2.1 Reagents. All chemicals used shall be reagent grade, or as specified below.

a. 5% silver nitrate: dissolve 5.0 g of silver nitrate in 100 mL water
b. Nitric acid, reagent grade, concentrated 67-70% wt
c. Potassium chloride, 5 mg/L stock solution (5 mg/L KCl): In a 1 L volumetric flask, dissolve 100 ± 1 mg potassium chloride in water, make up to volume and mix. Further, dilute a 5.00 mL aliquot of this solution to 100 mL with water.

4.4.6.2.2 Equipment.

a. All glassware shall be made free from soluble chlorides prior to use by rinsing with water.
b. 50 mL Nessler tubes

4.4.6.2.3 Sample preparation. In a 100 mL volumetric flask dissolve 1.00 ± 0.01 g of the potassium sulfate sample in water, dilute to volume and mix. If necessary, filter through chloride-free Whatman No. 40 filter paper prior to use.

4.4.6.2.4 Procedure. To a set of four (4) matched Nessler tubes add 20 mL water, and 1 mL each of nitric acid and 5% silver nitrate solution. To each respective tube add aliquots of sample solution and of the 5 mg/L KCl stock specified in Table V, below. Make up the Nessler tubes to volume with water, mix, and set in the dark for ten minutes to allow turbidity to develop.

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TABLE V. Nessler tube concentrations.

Tube designation	Volume of sample solution, mL	Volume of 5 mg/L KCl, mL
Blank	-	-
Reference, 0.01% wt KCl (1/2 of spec. max)	-	5.0
Reference, 0.02% wt KCl (spec. max)	-	10.0
Sample	25	-

4.4.6.2.5 Evaluation. Examine the relative opacities of the solutions by looking lengthwise through the Nessler tubes against a dark (black or brown) matte background; allow for adequate lighting but avoid bright lights and reflections. The blank preparation shall not present any visible turbidity. If it does, the above preparations shall be repeated with chloride-free water after ensuring that all glassware has been thoroughly cleaned of soluble chlorides and fine particulates. If the sample is unambiguously more opaque than the 0.02% KCl reference suspension, the sample fails the chlorides requirement. If the sample is clearly less opaque than the 0.01% KCl reference suspension, the sample passes chlorides requirement. If the sample opacity falls between those of the 0.01% KCl and 0.02% KCl, or if there is any uncertainty in this observation, then the primary, gravimetric method shall be used.

4.4.7 Determination of granulation. An accurately weighed portion of 100.0 ± 0.1 g of potassium sulfate shall be placed on the specified nest of sieves complying with ASTM E11, properly arranged, and assembled with a bottom pan. Two metal washers, weighing not more than 15 g each, shall be added to the upper sieve to help break up lumps. The nested sieves shall be covered and shaken for 5 minutes by hand or mechanically by means of a shaker geared to produce 285-315 gyrations and 150 ± 10 taps of the striker per minute. If necessary, the material on each sieve, beginning with the top one, shall be brushed lightly with a ½ inch flat bristle brush until no further material passes through each sieve. The portions retained on the various sieves and bottom pan shall be weighed (to the nearest 0.1 mg), and percent passing or retained shall be calculated on the basis of total sample recovered as follows:

$$\text{Percent through designated sieve} = 100\% * \frac{S}{\mathrm{W}}$$

$$\text{Percent retained on designated sieve} = 100\% * \frac{R}{\mathrm{W}}$$

where:

S = sum of weights of sample recovered from sieves below designated sieve and from bottom pan, g
R = weight of sample recovered from designated sieve, g
W = total weight of sample recovered from all sieves and bottom pan, g

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4.4.8 Determination of average particle size. The average particle size of Type III only shall be determined on a Fisher Sub-Sieve Sizer or equivalent instrument. The procedure is described in MIL-STD-650 (method 205.1) and in ASTM B330.

4.4.9 Workmanship. Take 200 ± 20 g of the material and spread the composition out on a clean white surface. The sample shall be visually examined, free from lumps, visible impurities, foreign matter, or other defects. Failure to meet the workmanship inspection or during following production processes shall result in batch or lot rejection.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DOD personnel, or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. Potassium sulfate covered by this specification is intended for use in ordnance applications.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type of material required (see 1.2).
c. Packaging requirements (see 5.1).
d. Requirements for first article samples (see 4.2).
e. Provision for submission of first article samples.
f. Requirements for submission of acceptance inspection equipment.

6.3 Technical agency. US Army Combat Capability Development Command Armaments Center (DEVCOM AC) is the technical agency that is responsible for the preparation and revision of the specification. All correspondence to the Technical Agency should be submitted through the contracting officer to: DEVCOM AC, ATTN: FCDD-ACE-QEE, Picatinny Arsenal, NJ 07806-5000.

6.4 Approval of equivalent test methods. Prior approval by the Contracting Officer should be required for use of equivalent test methods and/or instruments. A description of the proposed method and/or instrumentation should be submitted through the Contracting Officer to the Technical Agency (see 6.3). This description should include but not be limited to the procedures used, the accuracy and

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precision of the method, test data to demonstrate the accuracy and precision, and drawings of any special equipment required.

6.5 Submission of alternative conformance inspection provisions. Unless otherwise specified in the contract, proposed alternative conformance inspection provisions should be submitted by the contractor for evaluation by the Technical Agency (See 6.3) responsible for preparation of this specification.

6.6 Incorporated changes. This specification was updated from MIL-DTL-193B to MIL-DTL-193C as part of the Munitions Chemical Specification Modernization (MCSM) program. All test procedures were verified and updated as needed to produce valid results. Reference documents have been verified, updated, and hyperlinked to sources. Sampling procedures have been updated.

Specification ID	ECP	NOR
MIL-P-193A w/ Amendment 1	-	-
MIL-DTL-193B	Y23Q2007	1
MIL-DTL-193C	Y25Q2001	1

6.7 Subject term (key word) listing.

Propellant
Munitions Chemical Specification Modernization (MCSM)

6.8 Amendment notations. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Army – AR
Navy – OS
Air Force – 70

Preparing activity:
Army – AR
(Project 6810-2024-015)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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